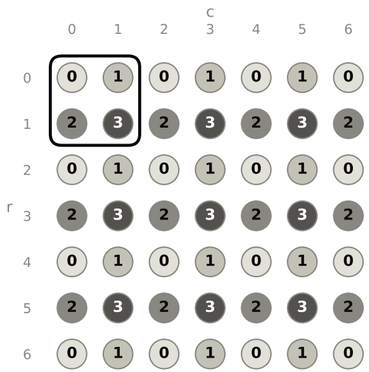
R1 code names: data qubit (c, r) takes letter k of the name's suffix (k = 0 is the first letter)

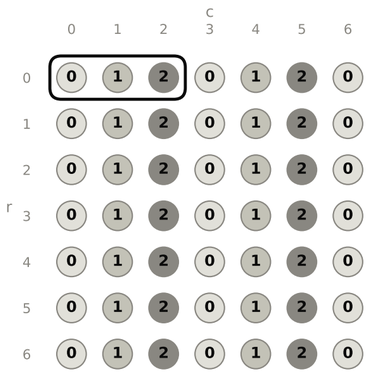
(c, r) = (0, 0) is the data qubit at (1, 1), top left; x = 2c + 1 grows to the right and y = 2r + 1 grows down, as in stim's diagrams

cell: k = 2 (r mod 2) + (c mod 2)



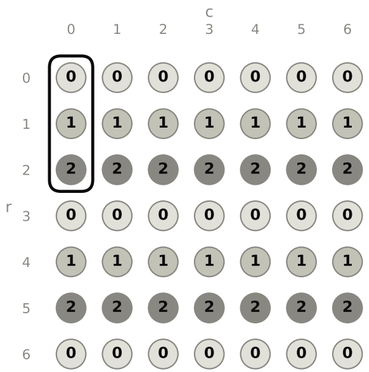
2×2 unit cell: top row, then bottom row

row3: k = c mod 3



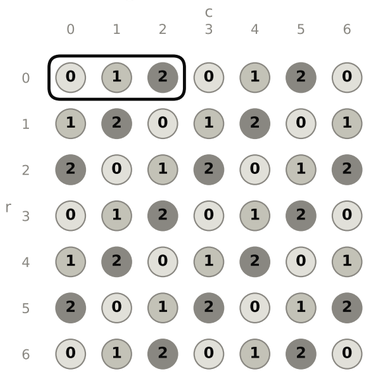
period 3 along a row: columns are constant

col3: k = r mod 3



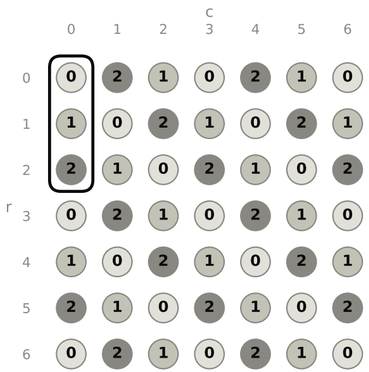
period 3 along a column: rows are constant

diag3: k = (r + c) mod 3



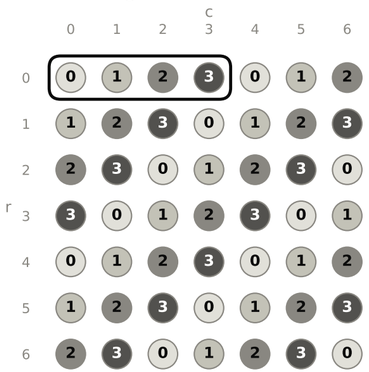
constant along every / diagonal

anti3: k = (r − c) mod 3



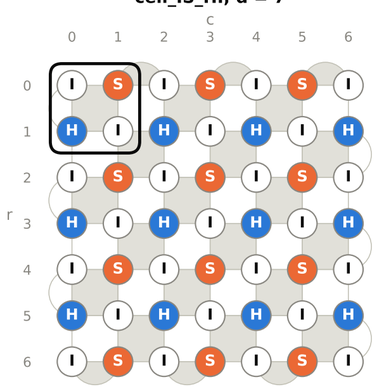
constant along every \ diagonal

diag4: k = (r + c) mod 4



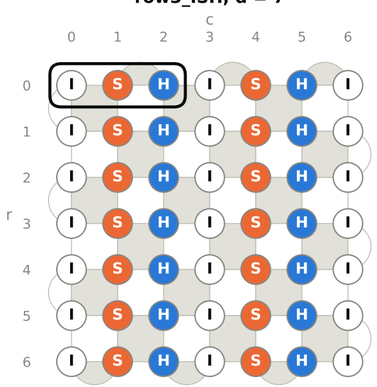
constant along every / diagonal

cell_IS_HI, d = 7



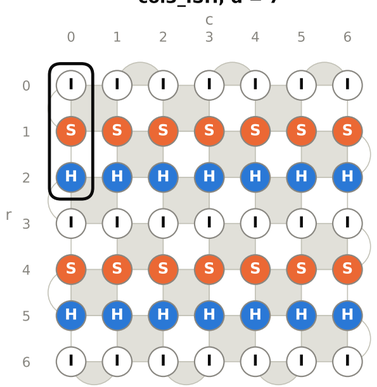
k = 0 → I, 1 → S, 2 → H, 3 → I

row3_ISH, d = 7



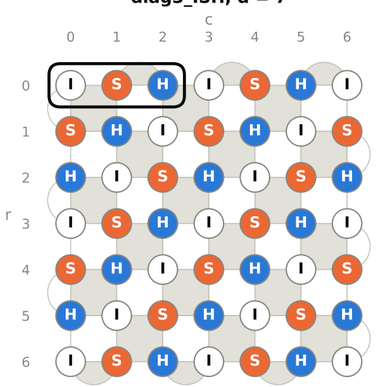
k = 0 → I, 1 → S, 2 → H

col3_ISH, d = 7



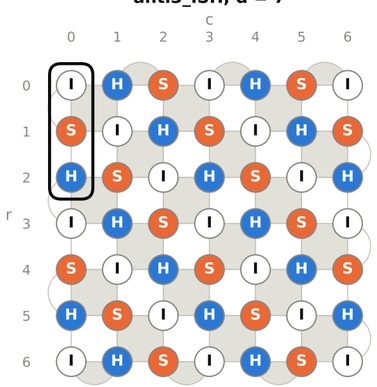
k = 0 → I, 1 → S, 2 → H

diag3_ISH, d = 7



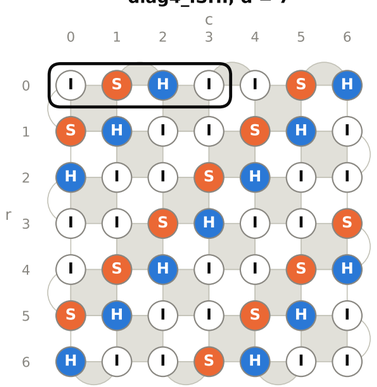
k = 0 → I, 1 → S, 2 → H

anti3_ISH, d = 7



k = 0 → I, 1 → S, 2 → H

diag4_ISHI, d = 7



k = 0 → I, 1 → S, 2 → H, 3 → I

I: none; a Z error stays a Z error H: Hadamard; a Z error acts as X on the CSS code S: √X; a Z error acts as Y on the CSS code Z check of the CSS code X check of the CSS code the qubits that spell the suffix in order